

A linear vector duality theory for decision support in joint venture planning

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ABSTRACT

Duality for linear vector optimum problems is often just a mathematical construct (l'art pour l'art). We define a family of dual problems to support decision-makers in a concrete economic situation. Entrepreneurs generally want to participate in initiatives for producing new products with similar technologies in a unique production setting. Each of them provides resources relative to the others' consideration of their economic objective. They all agree to aggregate their different interests through a new duality concept. This paper first presents the duality concept and shows under which conditions a satisfying aggregation of the entrepreneurs' varying objectives is possible. Numerical examples are used to make this aggregation transparent. An application is then presented to demonstrate the applicability and efficacy of the new concept.

1. Introduction

The field of vector or multi-objective optimization has experienced a boom in research interest in recent years, as Fig. 1 illustrates. In total, a search on the Web of Science platform yields over 5000 peer-reviewed articles starting in 1980, using the three terms “vector optimization”, “multi-objective optimization”, and “multiple objective optimization”. These contributions open up a wide range of applications such as production planning, financial planning, aerospace engineering, educational planning, infrastructure management, ecological modelling, disease analyses etc.; see, e.g., [1–13]. This emerging trend towards vector optimization is due to the fact that our modern society no longer wants decision problems to be reduced to a one-dimensional view, but rather the simultaneous consideration of multiple facets of an issue is the new credo.

It is surprising, however, that the term duality (theory) can be found in only 115 of this huge set of manuscripts. And more importantly, none of these manuscripts uses duality theory to rethink data structures in the context of joint venture planning, which is what this paper is about.

Duality for classical linear programming was one of the highlights in the field of Operations Research (OR) already in the 50s of the last century. The primal problem joins economic activities $\mathbf{x}$ and a technology matrix $\mathbf{A}$, a resources vector $\mathbf{b}$, and vector $\mathbf{c}$ of the activities' yield or profit. The dual problem then allows the calculation of the resources' (dual) prices. All this we find in early textbooks like [14–16], already. Very soon, the OR community criticized the single objective concept and started considering Linear Vector Optimization Problems (LVOPs); cf. [17–19]. Now, it needed a new comprehension of optimality for the vector-valued objective. Consequently, it needed a new duality concept

or – better – duality concepts. We name early works of Schöfeld [20], Kuhn and Tucker [21], Kornbluth [22], Isermann [23], Rödder [24] and Rödder and Leberling [25]. The latter is a working paper and therefore not available to a wider audience. A short, but incomplete introduction the reader also finds in [26]. To make this paper self-contained, we repeat some basic ideas from [25,26]. Recent theoretical work on duality theories and corresponding embeddings or connections between them can be found, for example, in [27–33].

This contribution is organized as follows: Section 2 gives preliminaries. In Section 3.1, we define a family of dual LVOPs and study their properties. Section 3.2 is central in as much it treats the question of a satisfying aggregation of the LVOPs' objectives. Illustrative numerical examples complement this elaboration. Section 4 prepares the economical point of view and gives dimensions of all parameters and variables in the dual models. Section 5 shows the capability of the new theory for application in the context of joint ventures. Section 6 is a summary and sketches possible future work.

2. Preliminaries

This section is a must to give all symbols and relations a clear meaning in the remainder of this contribution. Therefore, let $\mathbb{R}^d$ be the d -dimensional vector space. For $\mathbf{x}^T = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $\mathbf{y}^T = (y_1, \dots, y_d) \in \mathbb{R}^d$, we have

$$\begin{aligned} \mathbf{x} &\geq \mathbf{y} && \text{iff } x_i \geq y_i \quad \forall i \in \{1, \dots, d\}, \\ \mathbf{x} &\geq \mathbf{y} && \text{iff } \mathbf{x} \geq \mathbf{y} \text{ and } \mathbf{x} \neq \mathbf{y}, \\ \mathbf{x} &> \mathbf{y} && \text{iff } x_i > y_i \quad \forall i \in \{1, \dots, d\}, \end{aligned}$$

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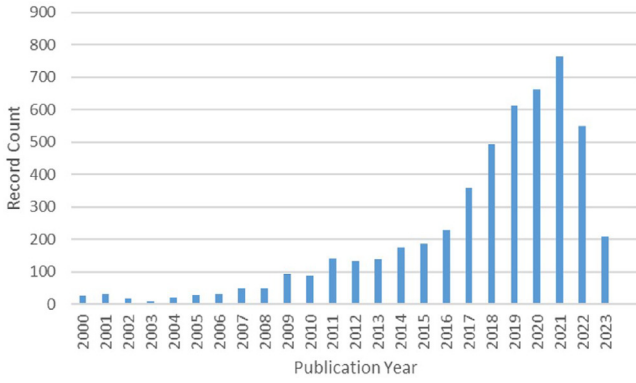


Fig. 1. Number of publications per year.

$\mathbf{x} \leq (\leq, <) \mathbf{y}$ iff $-\mathbf{x} \geq (\geq, >) -\mathbf{y}$.

If $\geq (\geq, >)$ is not true, we write $\nless (\nless, \nless)$, for short.

For $Z \subseteq \mathbb{R}^d$, we call $\mathbf{z}^\circ \in Z$ efficient in Z iff $\forall \mathbf{z} \in Z$ ($\mathbf{z} \geq \mathbf{z}^\circ \Rightarrow \mathbf{z} = \mathbf{z}^\circ$). If F is a mapping $\mathbb{R}^d \supseteq X \xrightarrow{F} \mathbb{R}^d$ and $Z = F(X)$, then $\mathbf{x}^\circ \in X$ is called functional efficient (f.e.) for F in X iff $F(\mathbf{x}^\circ)$ is efficient in Z . ‘Max’ $_{\mathbf{x} \in X} F(\mathbf{x})$ is the task to find all f.e. solutions for F in X , ‘Min’ $_{\mathbf{x} \in X} F(\mathbf{x})$ is equivalent to ‘Max’ $_{\mathbf{x} \in X} -F(\mathbf{x})$.

For the linear vector maximum problem, we use classical indexation and write

$$\text{‘Max’ } \mathbf{C}^T \mathbf{x} \quad \text{s.t. } \mathbf{A} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0} \quad (1)$$

($\mathbf{C}^T(K \times n)$, $\mathbf{A}(m \times n)$, $\mathbf{x}$, $\mathbf{0} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$).

Lemma 1. $\mathbf{x}^\circ$ is a f.e. solution iff there exists a $\mathbf{y} \in \mathbb{R}^K, \mathbf{y} > \mathbf{0}$ such that $\mathbf{x}^\circ$ solves

$$\text{Max } \mathbf{y}^T \mathbf{C}^T \mathbf{x} \quad \text{s.t. } \mathbf{A} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}. \quad (2)$$

For a proof of Lemma 1, see [23] and [34], Theorem 2.11. Finding all f.e. solutions for this convex problem (1) is equivalent to finding all solutions of the parametric problem (2). We mention the obvious fact that Lemma 1 also holds with the additional restriction $\sum_{k=1}^K y_k = 1$.

3. A duality theory for LVOPs

3.1. The theory and basic results

Now, we are ready to study a special duality concept for LVOPs. In so doing, we define the family of pairs of equations

$$\text{‘Max’ } \mathbf{C}^T \mathbf{x} \quad \text{s.t. } \mathbf{A} \mathbf{x} \leq \mathbf{B} \mathbf{y}, \mathbf{x} \geq \mathbf{0} \quad (3a)$$

$$\text{‘Min’ } \mathbf{u}^T \mathbf{B} \quad \text{s.t. } \mathbf{u}^T \mathbf{A} \geq \mathbf{y}^T \mathbf{C}^T, \mathbf{u} \geq \mathbf{0} \quad (3b)$$

($\mathbf{x} \in \mathbb{R}^n$, $\mathbf{u} \in \mathbb{R}^m$, $\mathbf{y} \in \mathbb{R}^K$, $\mathbf{C}^T(K \times n)$, $\mathbf{A}(m \times n)$, $\mathbf{B}(m \times K)$). Here and in the remainder of this paper, $\mathbf{y}$ is a common parameter for (3a) and (3b), with $\mathbf{y} > \mathbf{0}$, $\sum_{k=1}^K y_k = 1$.

Definition 1. For a fix $\mathbf{y} = \bar{\mathbf{y}}$, Eqs. (3a) and (3b) are a pair of dual LVOPs, with (3a) being the primal and (3b) the dual one.

$\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}^T$ are fix matrices, $\mathbf{y}$ is a parameter and $\mathbf{x}$, $\mathbf{u}$ are variables. The following Lemma 2 is a first justification for naming Eqs. (3a) and (3b) dual LVOPs.

Lemma 2.

- (i) If both Eqs. (3a) and (3b) are feasible for $\mathbf{y} = \bar{\mathbf{y}}$, then $\mathbf{C}^T \mathbf{x} \nless (\mathbf{u}^T \mathbf{B})^T \forall \mathbf{x}, \mathbf{u}$.

- (ii) If $\mathbf{C}^T \bar{\mathbf{x}} = (\bar{\mathbf{u}}^T \mathbf{B})^T$ for some feasible $\bar{\mathbf{x}}, \bar{\mathbf{u}}$, then $\bar{\mathbf{x}}$ is f.e. for (3a) and $\bar{\mathbf{u}}$ is f.e. for (3b).

Proof.

- (i) Assuming $\mathbf{C}^T \mathbf{x} \geq (\mathbf{u}^T \mathbf{B})^T$ for some $\mathbf{x}, \mathbf{u}$ implies $\bar{\mathbf{y}}^T \mathbf{C}^T \mathbf{x} > \mathbf{u}^T \mathbf{B} \bar{\mathbf{y}}$. On the other hand

$$\text{Max } \bar{\mathbf{y}}^T \mathbf{C}^T \mathbf{x} \quad \text{s.t. } \mathbf{A} \mathbf{x} \leq \mathbf{B} \bar{\mathbf{y}}, \mathbf{x} \geq \mathbf{0}$$

and

$$\text{Min } \mathbf{u}^T \mathbf{B} \bar{\mathbf{y}} \quad \text{s.t. } \mathbf{u}^T \mathbf{A} \geq \bar{\mathbf{y}}^T \mathbf{C}^T, \mathbf{u} \geq \mathbf{0}$$

are classical feasible linear problems (LPs) with $\bar{\mathbf{y}}^T \mathbf{C}^T \mathbf{x} \leq \mathbf{u}^T \mathbf{A} \mathbf{x} \leq \mathbf{u}^T \mathbf{B} \bar{\mathbf{y}}$, contradicting the assumption.

- (ii) Assuming $\mathbf{C}^T \bar{\mathbf{x}}$ not f.e. for (3a) implies the existence of an $\bar{\mathbf{x}}$ and $\mathbf{C}^T \bar{\mathbf{x}} \geq \mathbf{C}^T \bar{\mathbf{x}} = (\bar{\mathbf{u}}^T \mathbf{B})^T$, contradicting (i).

Assuming $\bar{\mathbf{u}}^T \mathbf{B}$ not f.e. for (3b) implies the same contradiction. $\square$

Lemma 2 impedes the strict dominance of the primal over the dual function value. Furthermore, equality of such values yields functional efficiency for both. That is what duality should do. But there is still an open question whether equality $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$ always is possible. It is not always possible but yet under weak conditions. More on that in the following section.

3.2. Duality and equality $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$

Motzkin in 1936 published his PhD thesis “Beiträge zur Theorie der linearen Ungleichungen” [35]. For a presentation easier accessible, cf. [19], p. 28. We put Motzkin’s result as a theorem, using the notation as in our LVOP context.

Theorem 1. Eq. (4)

$$\begin{aligned} 0\mathbf{x} + 0\mathbf{u} + \mathbf{I}\mathbf{y} &> \mathbf{0} \\ -\mathbf{A}\mathbf{x} + 0\mathbf{u} + \mathbf{B}\mathbf{y} &\geq \mathbf{0} \\ \mathbf{I}\mathbf{x} + 0\mathbf{u} + 0\mathbf{y} &\geq \mathbf{0} \\ 0\mathbf{x} + \mathbf{A}^T\mathbf{u} - \mathbf{C}\mathbf{y} &\geq \mathbf{0} \\ 0\mathbf{x} + \mathbf{I}\mathbf{u} + 0\mathbf{y} &\geq \mathbf{0} \\ \mathbf{C}^T\mathbf{x} - \mathbf{B}^T\mathbf{u} + 0\mathbf{y} &= \mathbf{0} \end{aligned} \quad (4)$$

or Eq. (5)

$$\begin{aligned} 0\mathbf{z}^1 + (-\mathbf{A}^T, \mathbf{I}, 0, 0)\mathbf{z}^2 + \mathbf{C}\mathbf{z}^3 &= \mathbf{0} \\ 0\mathbf{z}^1 + (0, 0, \mathbf{A}, \mathbf{I})\mathbf{z}^2 - \mathbf{B}\mathbf{z}^3 &= \mathbf{0} \\ \mathbf{I}\mathbf{z}^1 + (\mathbf{B}^T, 0, -\mathbf{C}^T, 0)\mathbf{z}^2 + 0\mathbf{z}^3 &= \mathbf{0} \\ \mathbf{z}^1 \geq \mathbf{0}, \mathbf{z}^2 \geq \mathbf{0}, \mathbf{z}^3 \text{ arbitrary} \end{aligned} \quad (5)$$

has a solution, but never both.

We observe that Eq. (4) is the Eqs. (3a), (3b) including $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$. After renaming variables and adequate transpositions, Eq. (5) reads

$$\mathbf{u}^T \mathbf{A} \geq \mathbf{y}^T \mathbf{C}^T, \mathbf{A} \mathbf{x} \leq \mathbf{B} \mathbf{y}, \mathbf{C}^T \mathbf{x} \geq (\mathbf{u}^T \mathbf{B})^T, \mathbf{x} \geq \mathbf{0}, \mathbf{u} \geq \mathbf{0} \text{ and } \mathbf{y} \text{ arbitrary!} \quad (5')$$

The solution of Eq. (4) meets mathematical rationale: There exist a positive $\bar{\mathbf{y}} > \mathbf{0}$ and a feasible $\bar{\mathbf{x}}, \bar{\mathbf{u}}$ for Eqs. (3a) and (3b) with equal primal and dual objective function values. If Eq. (4) has no solution, then Eq. (5) does. Eq. (5) is against mathematical rationale, as the following lemma illustrates.

Lemma 3. Each solution of Eq. (5') necessarily implies $y_k \leq 0$ for some k .

Proof. From Lemma 2 (i) we know that for feasible solutions $\mathbf{x}$ and $\mathbf{u}$ in the dual pair (3a) and (3b) and $\mathbf{y} = \bar{\mathbf{y}}$ always $\mathbf{C}^T \mathbf{x} \not\geq (\mathbf{u}^T \mathbf{B})^T$ holds. So in Eq. (5') $y_k \leq 0$ for some k . $\square$

For $y_k = 0$, the k th objective function does not exist in the primal LVOP. $y_k < 0$, for some k , is a severe aberration from our duality concept. The consequences of such aberration will be elaborated in the application section. We mention again the fact that all observations in this section remain true with the additional constraint $\sum_{k=1}^K y_k = 1$.

Please note: The non-existence of a solution of Eq. (4) is not a flaw of the duality concept but rather a flaw of the choice of the data $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}^T$.

Next, we show little numerical examples justifying the following statements: There exist data $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}^T$ for which both

- (3a) and (3b) are infeasible;
- (3a) and (3b) are feasible and $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$ for all f.e. $\mathbf{x}$, $\mathbf{u}$;
- (3a) and (3b) are feasible and $\mathbf{C}^T \mathbf{x} \neq (\mathbf{u}^T \mathbf{B})^T$ for all f.e. $\mathbf{x}$, $\mathbf{u}$ and hence Eq. (5') holds;
- (3a) and (3b) are feasible and $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$ for some f.e. $\mathbf{x}$, $\mathbf{u}$ and $\mathbf{C}^T \mathbf{x} \neq (\mathbf{u}^T \mathbf{B})^T$ for other f.e. $\mathbf{x}$, $\mathbf{u}$.

Now we provide the set of LVOPs like in (3a) and (3b) with the characteristics mentioned before.

(i)

$$\begin{aligned} \text{'Max'} \quad & \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{'Min'} \quad & (u_1, u_2) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \\ \text{s.t.} \quad & (u_1, u_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \geq (y_1, y_2) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

$$\mathbf{x}^T = (x_1, x_2) \geq \mathbf{0}; \mathbf{u}^T = (u_1, u_2) \geq \mathbf{0}; \mathbf{y}^T = (y_1, y_2) > \mathbf{0}; y_1 + y_2 = 1.$$

Both LVOPs are dual but infeasible for all $\mathbf{y}$.

(ii)

$$\begin{aligned} \text{'Max'} \quad & \begin{pmatrix} 1 \\ 1 \end{pmatrix} x \\ \text{s.t.} \quad & 1 \cdot x \leq (1, 1) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{'Min'} \quad & u(1, 1) \\ \text{s.t.} \quad & u \cdot 1 \geq (y_1, y_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

$$x \geq 0; u \geq 0; \mathbf{y}^T = (y_1, y_2) > \mathbf{0}; y_1 + y_2 = 1.$$

Both LVOPs are dual and feasible. For all $\mathbf{y}$, equality holds:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} x = (u(1, 1))^T \text{ for } x = 1, u = 1.$$

(iii)

$$\begin{aligned} \text{'Max'} \quad & \begin{pmatrix} 1 \\ 2 \end{pmatrix} x \\ \text{s.t.} \quad & 1 \cdot x \leq (1, 1) \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{'Min'} \quad & u(1, 1) \\ \text{s.t.} \quad & u \cdot 1 \geq (y_1, y_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

$$x \geq 0; u \geq 0; \mathbf{y}^T = (y_1, y_2) > \mathbf{0}; y_1 + y_2 = 1.$$

Both LVOPs are dual for all $\mathbf{y}$ but for f.e. $x = 1$ and $u = (y_1, y_2) \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ equality of primal and dual function values fails. Following the reasoning in Section 3.2, this implies in Eq. (5'):

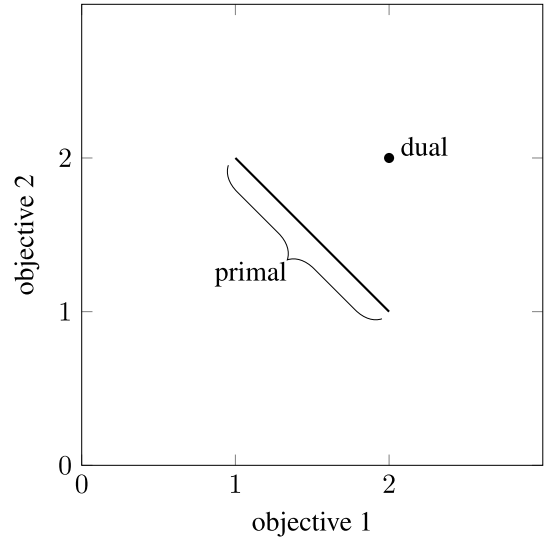


Fig. 2. Primal and dual f.e. objective function values, $y_1 \cong 1$.

$$\exists(x, u, \mathbf{y}) : u\mathbf{A} \geq (y_1, y_2)\mathbf{C}^T, \mathbf{A}\mathbf{x} \leq \mathbf{B} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \mathbf{C}^T \mathbf{x} \geq (u\mathbf{B})^T; \text{ choose } x = 1, u = 3/4, y_1 = 5/4, y_2 = -1/4.$$

(iv)

$$\begin{aligned} \text{'Max'} \quad & \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{'Min'} \quad & (u_1, u_2) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ \text{s.t.} \quad & (u_1, u_2) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \geq (y_1, y_2) \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \end{aligned}$$

$$\mathbf{x}^T = (x_1, x_2) \geq \mathbf{0}; \mathbf{u}^T = (u_1, u_2) \geq \mathbf{0}; \mathbf{y}^T = (y_1, y_2) > \mathbf{0}; y_1 + y_2 = 1.$$

For $y_1 \cong 1$, we get for f.e. $\mathbf{x} : x_1 + x_2 = 1$ and hence all convex combinations of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ for efficient objective function values.

For $y_1 \cong 1$, we get for f.e. $\mathbf{u} : u_1 + u_2 \cong 2$. Efficient value of $(u_1, u_2) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ then is $\begin{pmatrix} \approx 2 \\ \approx 2 \end{pmatrix}$.

The choice of $y_1 = y_2 = 1/2$, however, does not change the situation for the primal but makes efficient value of the dual objective value equal $(3/2, 3/2)$.

Resuming: There is no equality in the case $y_1 \cong 1$, but equality for $y_1 = y_2 = 1/2$. Reachability of equality depends on parameter $\mathbf{y}$. Figs. 2 and 3 might make this result more transparent.

The right choice of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}^T$ is a challenging task for a decision maker, using the presented duality concept. The next section paves the way from a mathematical to an economical concept.

4. From mathematical towards economical LVOPs

As announced in the introduction, entrepreneurs want to merge production activities in a new factory. They plan common activities $\mathbf{x}^T = (x_1, \dots, x_j, \dots, x_n)$, with $\dim(x_j)$ being the dimension or unit of x_j of product P_j . $\mathbf{A} = (a_{ij})_{m,n}$ is the technology matrix and m the number of resources, with $re_i, i = 1, \dots, m$, being their dimensions. Hence, $\dim(a_{ij}) = re_i/x_j$. $\mathbf{C}^T = (c_{kj})_{K,n}$; $j = 1, \dots, n$ and $k = 1, \dots, K$ are elements of the K linear objective functions, with o_k being their dimensions. Hence, $\dim(c_{kj}) = o_k/x_j$. As consideration of each stakeholder has to be

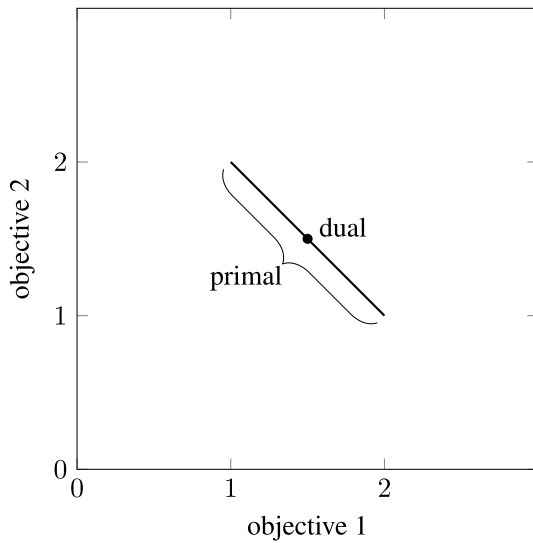


Fig. 3. Primal and dual f.e. objective function values, $y_1 = y_2 = 1/2$.

negotiated they agree to subordinate their respective interests under a common utility U , which in turn makes $\dim(y_k) = U/o_k$. This agreement causes a conditioned provision of resources for each of them. They offer resources relative to the importance of their objectives. Here matrix $\mathbf{B} = (b_{ik})_{m,K}$ comes into play, with $\dim(b_{ik}) = \frac{re_i}{U/o_k}$. When defining $\mathbf{u}^T = (u_1, \dots, u_i, \dots, u_m)$ with $\dim(u_i) = U/re_i$, we obtain $\mathbf{C}^T \mathbf{x} = (c_{kj}) \mathbf{x}$ with $\dim(\mathbf{C}^T \mathbf{x})_k = o_k$, $\mathbf{u}^T \mathbf{B} = \mathbf{u}^T (b_{ik})$ with $\dim(\mathbf{u}^T \mathbf{B})_k = o_k$. Ultimately, $\dim(\mathbf{A} \mathbf{x})_i = \dim(\mathbf{B} \mathbf{y})_i = re_i$; $\dim(\mathbf{u}^T \mathbf{A})_j = \dim(\mathbf{y}^T \mathbf{C}^T)_j = U/x_j$.

Consequently, all dimensions in Eqs. (4) and (5') are appropriate for respective calculations and relations. But are the decision-makers alias stakeholders able to calibrate the data $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}^T$ in such a way that a satisfying aggregation of their objectives $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$ is possible? Will the joint venture be successful?

5. A joint venture production of domestic appliances

Three entrepreneurs experienced in the production of domestic appliances want to join their activities to take advantage of scale effects. The first one up to now produced washing machines, the second one tumble dryers, and the third one dishwashers. As experts they know that production processes for these products are similar; and common use of a fabric, equipment, and man-power hopefully will benefit them all. In the initial meetings, it becomes clear that there are great difficulties in assessing each other's objective weighting and willingness to provide resources. They want to overcome this information deficit with the duality theory. Consequently, to help make decisions about planning shared resources, they want to use LVOP as they have different objective functions.

To keep this application transparent, we consider $\mathbf{x}^T = (x_1, x_2, x_3)$ annual production of washing machines, tumble dryers and dishwashers. All products run through the processes sheet metal forming, electric installation, assembly work, and finally packing and stocking. They

estimate the activity matrix $(a_{ij})_{4,3} = \begin{pmatrix} .2 & .2 & .3 \\ .1 & .3 & .15 \\ .3 & .2 & .2 \\ .2 & .3 & .1 \end{pmatrix}$ working hours for

product j in process i . The profits per unit of $\mathbf{x}^T = (x_1, x_2, x_3)$ are 100 €, 80 €, 90 €, respectively. In a first naïve agreement they want to share profits making their common vector-valued objective function

$\mathbf{C}^T = (c_{kj})_{3,3} = \begin{pmatrix} 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \end{pmatrix}$, giving an equal part of profit for each unit of the products to them all. Following the theoretical

discussion in the present paper, they opt for a conditioned provision of

annual resources $\mathbf{B} = (b_{ik})_{4,3} = \begin{pmatrix} 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \end{pmatrix}$ of stakeholder k in

working process i , $\frac{re_i}{U/o_k}$. Each of them apportions equal resources across the production processes. Entrepreneur $k = 1$ provides less resources as he/she feels aggrieved by the profit distribution. The reader might interpret the second and the third column of $\mathbf{B}$ in the same way.

All this now reads

$$\begin{aligned} \text{'Max' } \mathbf{C}^T \mathbf{x} &= \begin{pmatrix} 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\ \text{s.t. } \mathbf{A} \mathbf{x} &= \begin{pmatrix} .2 & .2 & .3 \\ .1 & .3 & .15 \\ .3 & .2 & .2 \\ .2 & .3 & .1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \mathbf{B} \mathbf{y} = \begin{pmatrix} 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{'Min' } \mathbf{u}^T \mathbf{B} &= (u_1, u_2, u_3, u_4) \begin{pmatrix} 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \\ 900 & 1125 & 1000 \end{pmatrix} \\ \text{s.t. } \mathbf{u}^T \mathbf{A} &= (u_1, u_2, u_3, u_4) \begin{pmatrix} .2 & .2 & .3 \\ .1 & .3 & .15 \\ .3 & .2 & .2 \\ .2 & .3 & .1 \end{pmatrix} \geq \mathbf{y}^T \mathbf{C}^T = \dots \\ & \quad (y_1, y_2, y_3) \begin{pmatrix} 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \\ 100/3 & 80/3 & 90/3 \end{pmatrix} \end{aligned}$$

$$\mathbf{x}^T = (x_1, x_2, x_3) \geq \mathbf{0}; \mathbf{u}^T = (u_1, u_2, u_3, u_4) \geq \mathbf{0}; \mathbf{y}^T = (y_1, y_2, y_3) > \mathbf{0}; y_1 + y_2 + y_3 = 1.$$

We observe that for all $\mathbf{x}$ and all $\mathbf{u}$, $\mathbf{C}^T \mathbf{x} = (\mathbf{u}^T \mathbf{B})^T$ never holds. There is no satisfying aggregation of primal and dual objectives. Due to a naïve data setting, the joint venture fails. Even worse: From Section 3 we know that the non-existence of a solution of Eq. (4) results in a solution of Eq. (5') with some non-positive y_k . We calculated the solution for (5') with data from this section. The result is

$$(x_1, x_2, x_3) \approx (321.29, 321.29, 8393.57)$$

$$(y_1, y_2, y_3) \approx (0, 1.77, -0.77)$$

$$(u_1, u_2, u_3, u_4) \approx (3.21, 3.21, 3.21, 3.21)$$

indicating an embarrassing affront of stakeholder no 3 against the others. He/she does not provide resources but plans a withdrawal. Furthermore, the imperative 'Max' $\mathbf{C}^T \mathbf{x}$ is in vain; the 3rd objective function is to be minimized instead of maximized, against mathematical and economical rationale – as emphasized in Section 3.2. That harms their common activities.

The sobering experience from a very naïve attitude makes the three stakeholders alter some of the data

- in that they make $\mathbf{C}^T = \begin{pmatrix} 100 & 0 & 0 \\ 0 & 80 & 0 \\ 0 & 0 & 90 \end{pmatrix}$ and
- in that they calibrate better their conditioned provisions of resources $\mathbf{B} = (b_{ik})_{4,3} = \begin{pmatrix} 2000 & 1500 & 3000 \\ 800 & 2000 & 1500 \\ 4000 & 1500 & 1500 \\ 2000 & 2000 & 1000 \end{pmatrix}$.

The entries of $\mathbf{C}^T$ now reflect the economic credo that each stakeholder receives the profit from his/her product exclusively. The entries in $\mathbf{B}$ now show a better adaption of resources' consumption re_i in respective

working processes. With these data, the dual LVOPs read

$$\begin{aligned} \text{'Max' } \mathbf{C}^T \mathbf{x} &= \begin{pmatrix} 100 & 0 & 0 \\ 0 & 80 & 0 \\ 0 & 0 & 90 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\ \text{s.t. } \mathbf{A} \mathbf{x} &= \begin{pmatrix} .2 & .2 & .3 \\ .1 & .3 & .15 \\ .3 & .2 & .2 \\ .2 & .3 & .1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \leq \mathbf{B} \mathbf{y} = \begin{pmatrix} 2000 & 1500 & 3000 \\ 800 & 2000 & 1500 \\ 4000 & 1500 & 1500 \\ 2000 & 2000 & 1000 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \\ \text{'Min' } \mathbf{u}^T \mathbf{B} &= (u_1, u_2, u_3, u_4) \begin{pmatrix} 2000 & 1500 & 3000 \\ 800 & 2000 & 1500 \\ 4000 & 1500 & 1500 \\ 2000 & 2000 & 1000 \end{pmatrix} \\ \text{s.t. } \mathbf{u}^T \mathbf{A} &= (u_1, u_2, u_3, u_4) \begin{pmatrix} .2 & .2 & .3 \\ .1 & .3 & .15 \\ .3 & .2 & .2 \\ .2 & .3 & .1 \end{pmatrix} \geq \mathbf{y}^T \mathbf{C}^T = \dots \\ & (y_1, y_2, y_3) \begin{pmatrix} 100 & 0 & 0 \\ 0 & 80 & 0 \\ 0 & 0 & 90 \end{pmatrix} \end{aligned}$$

$$\mathbf{x}^T = (x_1, x_2, x_3) \geq \mathbf{0}; \mathbf{u}^T = (u_1, u_2, u_3, u_4) \geq \mathbf{0}; \mathbf{y}^T = (y_1, y_2, y_3) > \mathbf{0}; y_1 + y_2 + y_3 = 1.$$

For these two problems, we solved Eq. (4); the results are

$$(x_1, x_2, x_3) \approx (1246.75, 3896.10, 2597.40)$$

$$(y_1, y_2, y_3) \approx (0.1558, 0.5844, 0.2597)$$

$$(u_1, u_2, u_3, u_4) \approx (0, 155.844, 0, 0)$$

$$\text{So } \mathbf{C}^T \mathbf{x} = \begin{pmatrix} 124,675 \\ 311,688 \\ 233,766 \end{pmatrix} \text{ is the expected profit } o_k/x_j \cdot x_j = o_k \text{ for the}$$

three stakeholders. What their respective investments is concerned we

$$\text{have } (\mathbf{u}^T \mathbf{B})^T = \begin{pmatrix} 124,675 \\ 311,688 \\ 233,766 \end{pmatrix}, U/re_i \cdot \frac{re_i}{U/o_k} = o_k. \text{ Both values coincide,}$$

as it should be. Another interesting information are the resources

$$\begin{pmatrix} 2000 \\ 800 \\ 4000 \\ 2000 \end{pmatrix} y_1 = \begin{pmatrix} 311.6 \\ 124.6 \\ 623.2 \\ 311.6 \end{pmatrix}; \begin{pmatrix} 1500 \\ 2000 \\ 1500 \\ 2000 \end{pmatrix} y_2 = \begin{pmatrix} 876.6 \\ 1108.8 \\ 876.6 \\ 1168.8 \end{pmatrix}; \begin{pmatrix} 3000 \\ 1500 \\ 1500 \\ 1000 \end{pmatrix} y_3 = \begin{pmatrix} 777.6 \\ 388.8 \\ 388.8 \\ 259.2 \end{pmatrix}. \text{ They}$$

must check whether they are able to provide such resources in addition to their still remaining activities, producing washing machines, tumble dryers, and dishwashers.

6. Conclusions

In Wikipedia, we find the definition: "In mathematics, a duality translates concepts, theorems or mathematical structures into other concepts, theorems or structures, ..." In the context of this contribution, the mathematical structures are a linear vector maximum problem and its dual linear vector minimum problem. Duality should enrich the knowledge about such mathematical structures. That is exactly what we observe for dual LVOPs.

We describe the planning process for a joint venture of entrepreneurs which want to organize common production in a new factory. In doing so, they must harmonize their (different) objectives and their (different) investments. Here, LVOP duality theory becomes effective. And this theory demands a set of compatible data like objectives, activities, and investments. In an application, incompatible data imply in a failure of the venture; compatible data end up in a successful venture. The achievement of compatible data is a difficult task in a situation with many entrepreneurs, activities and resources.

This data issue can also be linked to the topic of inverse problems as discussed by Ghate [36].

Is there a key for (iteratively) generating an economic expedient data set; and do these data reflect the willingness of actors to cooperate? Our future work must provide an answer to these questions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data are displayed in the main body of the manuscript.

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